

**Accuracy of therapeutic recommendation of a clinical decision support system:
Onechoice® report based on phenotypic results in relation to infectious disease
physicians and other specialties in the treatment of Urinary Tract Infection (UTI)
diagnosed in Lima-Peru**

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Key words: UTIs, Machine Learning.

Introduction: Urinary tract infections (UTI) are one of the most common pathologies in clinical practice and present challenges in treatment due to increasing antimicrobial resistance. The OneChoice® report is generated through a Human-in-the-loop (HITL) Machine Learning (ML)-based algorithm, which is based on demographic and phenotypic data received from laboratory results, as well as international guidelines (IDSA), scientific reviews, and approvals for the use of antimicrobials by agencies such as the FDA and CDC, as well as drug names and availability by country. This study evaluates the concordance between OneChoice® recommendations and the clinical decisions of specialists in Infectology and other specialties in Peru.

Materials and Methods: 100 clinical cases were designed (In Google Forms) from positive urine cultures reported in July 2024 in a private laboratory (Complete urine examination, germ isolated, colony count, and antibiograms), which had Onechoice® therapeutic recommendation and treatment alternatives based on HITL-ML, with questions focused on the best therapeutic recommendation and the alternative. The surveys were conducted by accessing an electronic link (random case selection algorithm) by different medical specialists. The specialists' decisions were compared with the OneChoice® recommendations, evaluating the accuracy and concordance of the medical specialties, in addition to considering the years of professional experience.

Results: Cases were resolved by Infectologists (33), Cardiologists (3), Endocrinologists (1), Rheumatologists (2), Geriatricians (4), Internal Medicine (33) and other specialties (59), with a total of 137 cases resolved, presenting a general concordance of 20%. In non-Infectious Disease physicians, the overall concordance (OneChoice®+Alternative) was 12%, OneChoice® concordance was 35% and the alternative was 29%. In the cases resolved by Infectologists (35), they had a global concordance of 43%, OneChoice® of 63% and the alternative of 49%. It was confirmed that the discrepancies caused were not due to a system error. In the concordances according to years of experience, it is evident that physicians with 1-5 years and those with more than 20 years of experience in their respective specialties are those who presented the best concordance both in OneChoice® and in the alternative to the treatment.

Conclusions and Recommendations: The specialty of infectious disease is the one that has the best concordance with the correct treatment and its respective alternative in the treatment of Urinary Tract Infections, being this lower in non-infectious disease doctors. These data reinforce the need to have systems based on HITL-ML that have been validated, in order to improve the therapeutic indications in urinary tract infections with positive urine cultures in Peru. It is recommended to increase the number of surveys for better analysis by specialty.

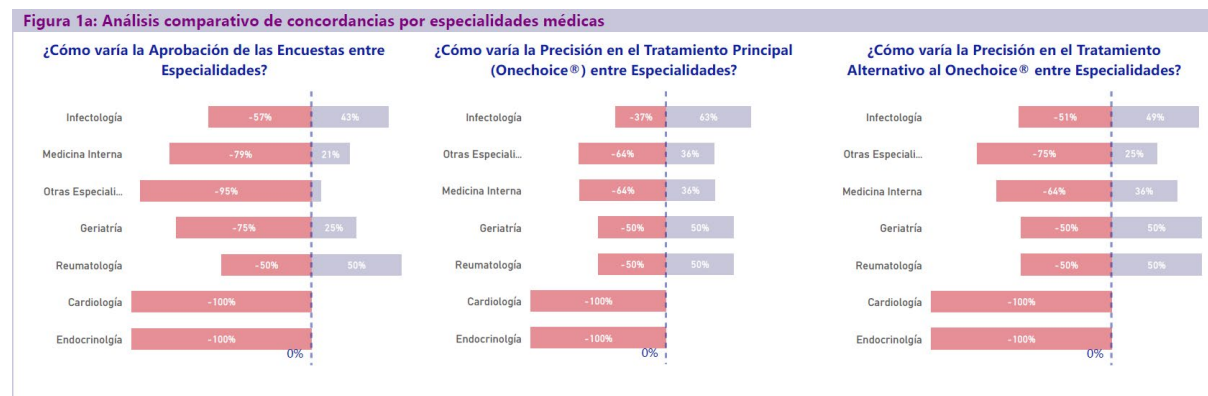


Figura 1b: Análisis comparativo de concordancias por especialidades médicas

